

4th Sem / Mechanical, Mech(Tool & Die)
Subject : Hydraulics and Pneumatics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 An ideal fluid is a fluid which is (CO1)

- a) Compressible
- b) Incompressible
- c) Incompressible and Viscous
- d) None of the above

Q.2 The Specific volume of a liquid is a reciprocal of (CO2)

- a) Weight density b) Mass density
- c) Specific weight d) Specific volume

Q.3 Manometer is used for measuring (CO3)

- a) Only high pressure
- b) Only low pressure
- c) Both high and low pressure
- d) Only negative pressure

Q.12 Hydraulic press is the practical application of _____
law. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any eight
questions out of ten questions. (8x4=32)

Q.13 Explain surface tension and capillarity. (CO1)

Q.14 State newton's law of viscosity. (CO1)

Q.15 If the specific gravity of a liquid is 13.6, find its mass
density and weight density. (CO2)

Q.16 Write a short note on compressibility. (CO2)

Q.17 Explain gauge pressure and absolute pressure.
(CO3)

Q.18 Enlist various faults in hydraulic systems.

Q.19 Explain pitot tube with the help of neat sketch.
(CO4)

Q.20 Explain loss of head of exit of pipe. (CO5)

Q.21 Derive an expression for the velocity of flow through
nozzle. (CO5)

Q.22 What is single acting air compressor? Explain with
the help of neat sketch.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Derive an expression for discharge measured by orifice meter. (CO4)

Q.24 Explain with neat sketch, construction and working of kaplan turbine. (CO5)

Q.25 What is hydraulic ram? Explain its construction and working with the help of neat sketch. (CO6)